PROOF OF BROSNAN'S p-INCOMPRESSIBLE CONJECTURE

JIHAO LIU

ABSTRACT. We prove Brosnan's p-incompressible conjecture: For any prime number p and any abelian variety, the multiplication by p map is p-incompressible.

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1. INTRODUCTION

Essential dimension, introduced in [BR97], measures the least number of independent parameters on which an algebraic object over a field depends. The essential dimension at a prime p [RY00] is the variant which ignores base changes of degree prime to p. For generically étale morphisms, both invariants are encoded in the notions of incompressibility and p-incompressibility of [FS22, §1], which we now describe.

Let k be a field and $f : X \to Y$ a morphism between varieties over k which is étale over a dense open subset of Y. Roughly speaking, f is *incompressible* if f is not obtained, over the generic point of Y and along a dominant rational map $Y \dashrightarrow Y'$, as the pullback of a morphism $f' : X' \to Y'$ which is étale over a dense open subset of a variety Y' with $\dim Y' < \dim Y$. For a prime p, the morphism f is *p-incompressible* if $X \times_Y Z \to Z$ is incompressible for any variety Z over k and any dominant generically finite morphism $Z \to Y$ of degree prime to p. We refer the reader to Definition 2.1 for the precise definitions.

Let A be an abelian variety over a field k of characteristic zero, $d = \dim A$, and p a prime. The multiplication by p map $[p] : A \to A$ is finite étale of degree p^{2d}, and the kernel $A[p]$ of $[p]$ is a finite group scheme over k whose points over an algebraic closure of k form a group isomorphic to $(\mathbb{Z}/p\mathbb{Z})^{2d}$, by [Mum74, §6, Application 3 and §7, Theorem 4, Corollary 1] applied after base change to an algebraic closure of k. When $k = \mathbb{C}$, we identify $A[p]$ with the group of p-torsion points of $A(\mathbb{C})$, which acts on A by translations, and $[p]$ is the quotient map of A by $A[p]$ [Mum74, §7, Theorem 4].

The following conjecture is due to Brosnan [FS22, Conjecture 6.1 and footnote 6].

Conjecture 1.1 ([FS22, Conjecture 6.1])**.** *Let A be an abelian variety over a field k of characteristic zero. Then for any prime p, the multiplication by p map $[p] : A \to A$ is p-incompressible.*

By [FS22, p. 177], Conjecture 1.1 implies that $[m] : A \to A$ is incompressible for any integer $m \geq 2$. The case $d = 1$ is trivial. When A is a very general product of elliptic curves,

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Conjecture 1.1 follows from [CT02, Appendice], according to [FKW24, Remark 2.3.6]. See also [FS22, p. 178].

Before this paper, the following results were known.

(1) [FS22, Corollary 6.5] proved Conjecture 1.1 when A has good reduction at a discrete valuation of k of residue characteristic p and the reduction has p-rank at least $d-1$. [FS22, Theorem 1.2] proved Conjecture 1.1 when $d \leq 3$ for a set of primes p of positive density depending on A.
(2) [FKW24, Corollary 2.3.5] proved Conjecture 1.1 when k is algebraically closed, $p \geq d+2$, and A has unramified good reduction at p in the sense of [FKW24, §2.2.12]. In particular, Conjecture 1.1 holds for p sufficiently large when k is algebraically closed. See also [FKW24, Remark 2.3.6].
(3) When k is algebraically closed, [KZ26, Theorem 1] proved that $[m]$ is incompressible for any integer $m \geq 2$, and the p-incompressibility of $[p]$ remains open [KZ26, end of §4].

[Che26, Remark 6.5.4] describes the status of Conjecture 1.1 in May 2026 and lists only the results in (1), (2), and (3).

In addition, [FS22, Theorem 6.4] proves that the essential dimension at p of $[p]$ is at least $n+1$ when the special fiber of the Néron model of A at a discrete valuation of k of residue characteristic p at which A has semi-stable reduction contains a closed subgroup scheme $(\mu_p)^n$ with $n < d$.

Even for $d = 2$, it was not known whether $[p]$ is p-incompressible for all primes p [FS22, Remark 6.8(1)]. For instance, let E be the elliptic curve over $\mathbb{Q}$ defined by $y^2 = x^3 - x$, let A be the base change of $E \times E$ to k, so that $d = 2$, and let $p = 3$. The discriminant of $x^3 - x$ is 4, so E has good reduction at 3. The reduction $E_{\mathbb{F}_3}$ of E modulo 3 satisfies

$$\#E_{\mathbb{F}_3}(\mathbb{F}_3) = \#\{(x, y) \in \mathbb{F}_3^2 \mid y^2 = x^3 - x\} + 1 = 3 + 1 = 4,$$

so $3 + 1 - \#E_{\mathbb{F}_3}(\mathbb{F}_3) = 0$ and $E_{\mathbb{F}_3}$ is supersingular. Thus at any discrete valuation of k of residue characteristic 3, A has good reduction and the reduction has 3-rank $0 < d - 1$. Therefore [FS22, Corollary 6.5] does not apply, [FS22, Theorem 1.2] does not apply since its proof uses [FS22, Corollary 6.5] at primes p at which the reduction has p-rank at least 2 [FS22, Lemma 6.6 and the proof of Corollary 6.7], [FS22, Theorem 6.4] only implies the trivial lower bound, [FKW24, Corollary 2.3.5] requires $p \geq d+2 = 4$, and [KZ26, Theorem 1] only proves incompressibility. Thus none of [FS22, Theorem 6.4] and the results in (1), (2), and (3) covers this case. Conjecture 1.1 was also open in general for abelian varieties A of dimension d and primes $p \geq d+2$ at which A has no unramified good reduction.

We now explain where the existing methods stop. The methods of [FS22] and [FKW24] use the reduction of A at p, and the method of [FS22] extends the mixed characteristic method of [FKW21] (see [FS22, p. 152]). The lower bound of [FS22, Theorem 6.4] is trivial when the reduction is supersingular. The lower bound of [FKW24, Corollary 2.3.5] relies on a non-vanishing which is proved only in degrees $i \leq p-2$ [FKW24, Corollary 2.2.13 and Proposition 2.3.3], which forces $p \geq d+2$, and the argument depends on unramified good reduction at p in the sense of [FKW24, §2.2.12].

In general, when A has good reduction at p and a compression of $[p]$ is spread out over the valuation ring, the action of the p-torsion subgroup scheme of the Néron model of A descends to the special fiber of the compressed variety, but the descended action is not known to be generically free (see [FS22, Remarks 2.3 and 6.8(2)]). In [FS22, Example 2.4], a subgroup scheme of order p^2 of the p-torsion of the square of the Néron model of an elliptic curve with good supersingular reduction acts freely on the square, and the action descended along the sum map is free on the generic fiber but not generically free on the special fiber.

The argument of [KZ26, §4] is birational. For an isogeny with source A, it starts from a rational map $A \dashrightarrow Y$ with connected fibers to which the action of the kernel of the isogeny descends, and then uses the Albanese map of Y, whose general fibers are rationally connected [KZ26, Corollary 7], together with the bound on the rank of abelian p-groups acting faithfully on

rationally connected varieties [KZ26, Corollary 11]. For p-incompressibility, one first passes to a dominant generically finite morphism $Z \to A$ of degree prime to p, and then has to compress $A \times_{[p],A} Z \to Z$. The variety $A \times_{[p],A} Z$ need not be an abelian variety, so in general the structure of rational images of abelian varieties [KZ26, Proposition 4] is not available, and $A \times_{[p],A} Z$ can be of general type, for example when $Z \to A$ is a cyclic cover of prime degree $m \neq p$ branched along a smooth divisor in $|mL|$ for an ample divisor L, since then K_Z is the pullback of the ample divisor $(m-1)L$ and $A \times_{[p],A} Z \to Z$ is étale. [KZ26, end of §4] notes that the method yields little information for morphisms between varieties of general type.

The fixed-point method [BF24, §1.2], whose general form is [BF24, Proposition 10(b)], proves that a torsor under a finite group is p-incompressible if a finite abelian p-subgroup of rank equal to the dimension of the total space has a smooth fixed point on some equivariant compactification of the total space. When $k = \mathbb{C}$, the total space of $[p]$ is A, which is proper, so the only equivariant compactification of A is A, and any non-zero element of $A[p]$ acts on A by a translation without fixed points. Hence the fixed-point method does not apply directly to $[p]$.

On the group level, [KM08, Theorem 4.1] computes the essential dimension at p of a finite p-group over a field of characteristic different from p containing a primitive p-th root of unity, and the essential dimension at p of $A[p] \cong (\mathbb{Z}/p\mathbb{Z})^{2d}$ over $\mathbb{C}$ is $2d$ [BF24, Remark 4]. Conjecture 1.1 considers one specific $A[p]$-torsor, namely $[p]$, not the group $A[p]$. For other self-maps, [KZ26, Question 19] asks whether any polarized endomorphism is incompressible. [LOZ25, Example 1.3] answers the question negatively on abelian surfaces, and [LZ26, Theorem 1.6] proves that any finite Galois self-cover of $\mathbb{P}^n$ of degree greater than 1 is incompressible.

Main results. The goal of this paper is to prove Conjecture 1.1.

Theorem 1.2. *Conjecture 1.1 holds.*

By Definition-Proposition 2.2, Theorem 1.2 is equivalent to the equality

$$\mathrm{ed}_k([p]; p) = \dim A$$

for any field k of characteristic zero, any abelian variety A over k, and any prime p, where $[p]$ denotes $[p] : A \to A$.

Theorem 1.2 is deduced in Section 4, by two reductions, from the following divisibility theorem over $\mathbb{C}$.

Theorem 1.3. *Let p be a prime, $d \geq 1$ and $0 \leq n \leq d$ integers, and A an abelian variety of dimension d. Let $G := A[p]$ act on A by translations. Let X and T be smooth projective varieties of dimensions d and n respectively, with regular G-actions. Let $q : X \to A$ be a G-equivariant dominant generically finite morphism of degree r, and $f : X \to T$ a G-equivariant surjective morphism. Assume that the G-action on T is faithful. Then p^{d-n} divides r.*

Sketch of the proof. Theorem 1.2 is proved by contradiction: by spreading out, Lemma 4.1 reduces a counterexample to one over $\mathbb{C}$, which Lemma 4.2 turns, using [AW97, Theorem 0.1], into the data of Theorem 1.3 with $n < d$ and $p \nmid r$, contradicting $p^{d-n} \mid r$. Theorem 1.3 follows from the holomorphic Atiyah–Bott formula [AB67, AB68], in the form of [KP20, Theorem 3.1.2] (Theorem 2.5), applied for each $h \in G$. Since $q(hx) = q(x) + h$ and a non-zero translation of A has no fixed points, the group G acts freely on X. Nevertheless, on a general fiber V of the sum map $X^p \to A$, $(x_i) \to \sum_i q(x_i)$, the composite $h\tau$ of h with the cyclic permutation τ of the factors of X^p has exactly $p^{2d}r$ fixed points, at each of which $\det(1 - \mathrm{d}(h\tau)) = p^d$ (Construction-Proposition 3.2 and Lemma 3.5). With coefficients in the derived pullback $\mathcal{E}$ to V of the structure sheaf of the small diagonal of T^p under $f \times \cdots \times f$ (Construction-Proposition 3.3), the local terms (Definition 2.4) are 0 for $h \neq 0$, since G acts faithfully on T, and p^n for $h = 0$ (Lemma 3.6). By projection to G-invariants, the alternating sum of the traces of τ on $H^j(V, \mathcal{E})^G$ is

$$\frac{1}{|G|} \cdot p^{2d} r \cdot \frac{p^n}{p^d} = \frac{r}{p^{d-n}}$$

(see (4.4)), which is rational and, since $\tau^p = 1$, an algebraic integer, hence an integer. The groups $H^j(V, \mathcal{E})^G$ are never computed. The new input is the choice of V, τ, and $\mathcal{E}$, after which Lemmas 3.5 and 3.6 are direct computations.

The proof of Theorem 1.3 uses no reduction modulo p, so the obstruction of [FS22, Remark 6.8(2)] does not arise. We now explain two new ideas and the reduction to equivariant data.

(a) Fixed points. Since G acts freely on X, the fixed-point method [BF24, §1.2] does not apply to X either. The cyclic permutation τ creates fixed points: the fixed locus of $h\tau$ on X^p is $\{(x, (-h)x, \dots, (-(p-1)h)x) \mid x \in X\} \cong X$. Since $ph = 0$, the sum map $s : X^p \to A$ is invariant under G and τ, and the restriction of s to the fixed locus of $h\tau$ is $[p] \circ q$ followed by a translation. Thus the fixed points of $h\tau$ on the fiber V of s over a general point correspond to the points of a general fiber of the generically finite morphism $[p] \circ q$. At such a point, the tangent space of V is identified, through $\mathrm{d}q$ and translations, with the space of vectors with zero sum in $(T_0 A)^{\oplus p}$, on which $\mathrm{d}(h\tau)$ acts by cyclic permutation, with eigenvalues $\zeta, \dots, \zeta^{p-1}$, each of multiplicity d, for a primitive p-th root of unity ζ, and $\prod_{j=1}^{p-1}(1 - \zeta^j) = p$.

Since G acts freely on X, hence on $V \subset X^p$, no point of V is fixed by a non-cyclic subgroup of $G \times \langle \tau \rangle$, so instead of the fixed-point method, we use the Lefschetz numbers of the single elements $h\tau$. In the topological Lefschetz fixed point formula, each fixed point of $h\tau$ has index 1, so the denominators p^d appear only in the holomorphic formula.

(b) The coefficient complex. On X itself, Theorem 2.5 yields 0 for any $h \neq 0$ and any coefficients, so the faithfulness of the G-action on T cannot enter. The complex $\mathcal{E}$ of Construction-Proposition 3.3 is supported on the preimage of the small diagonal $\Delta \subset T^p$. For $h \neq 0$, the choice of the general point in Construction-Proposition 3.2(3), the only place where the faithfulness is used, ensures that $f \times \cdots \times f$ maps the fixed points of $h\tau$ outside Δ, so the derived fibers of $\mathcal{E}$ at the fixed points of $h\tau$ vanish.

For $h = 0$, the derived fiber of $\mathcal{E}$ at a fixed point is the derived fiber of $\mathcal{O}_\Delta$ at a point of Δ, which does not depend on f, even where f is not smooth. Since T is smooth, $\mathcal{O}_\Delta$ is a perfect complex on T^p, and its τ-equivariant Koszul resolution shows that the local term is the determinant of $1 - \tau$ on the conormal space of Δ. With coefficients $\mathcal{O}_V$, any $h\tau$ contributes $p^{2d} r / p^d$, and the projection only yields the integer $p^d r$.

(c) Equivariant data from a compression. For $Z \to A$ of degree prime to p, a compression of $A \times_{[p],A} Z \to Z$ need not carry an action of G. Since $A \times_{[p],A} Z \to Z$ is a G-torsor, [BF24, Lemma 2], whose proof is omitted there, provides a G-equivariant compression after a finite étale cover of the base of the compression. **Step 3** of the proof of Lemma 4.2 constructs one directly: for a compression with base Z' of dimension less than d and with morphism $W' \to Z'$, it sends a point w of $A \times_{[p],A} Z$ over a general point of Z to the tuple of the images of hw, $h \in G$, in W'. The closure T_1 of the image is G-stable and finite over Z', so $\dim T_1 < d$, and G acts faithfully on T_1, since G acts freely on $A \times_{[p],A} Z$.

After the equivariant resolutions of [AW97, Theorem 0.1], T and X are smooth projective, X is birational to $A \times_{[p],A} Z$, which need not be an abelian variety (cf. [KZ26, end of §4]), and the action of G on T is faithful but need not be free. Theorem 1.3 assumes no more.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce incompressibility, p-incompressibility, and the essential dimension at p, and recall the holomorphic Atiyah–Bott formula for perfect complexes [KP20, Theorem 3.1.2].

In Section 3, in the setting of Theorem 1.3, we construct automorphisms of a general fiber of the sum map $X^p \to A$ with finitely many transverse fixed points, and compute the denominators and the local terms that enter the holomorphic Atiyah–Bott formula.

In Section 4 we prove Theorem 1.3, and then deduce Theorem 1.2 from it by two reductions: from a field of characteristic zero to $\mathbb{C}$, and from a compression of $[p]$ to the equivariant data of Theorem 1.3.

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2. Preliminaries

2.1. **Incompressibility.** The following definition is taken from [FS22, §1, pp. 151–152], written in the formulation of [BF24, §1.1] by pullbacks over the generic point. It is equivalent to the formulation of [FS22, §1] by Cartesian diagrams. Given a compression as in Definition 2.1 below, removing from X' the irreducible components which do not dominate Y' does not change the pullback over the generic point of Y, and gives a Cartesian diagram as in [FS22, §1]. Conversely, a Cartesian diagram as in [FS22, §1] gives, over the generic point of Y, a compression as in Definition 2.1, since étaleness over the generic point of the base descends along the extension of function fields.

Definition 2.1 ([FS22, §1])**.** Let k be a field. By a *variety over* k we mean an integral separated scheme of finite type over k. Let Y be a variety over k, and $f : X \to Y$ a separated morphism of finite type which is étale over a dense open subset of Y. A *compression* of f consists of a variety Y' over k, a dominant rational map $Y \dashrightarrow Y'$, and a separated morphism of finite type $f' : X' \to Y'$ which is étale over a dense open subset of Y', such that over the generic point of Y, the morphism f is the pullback of f'. We call Y' the *base* of the compression. We say that f is *incompressible* if $\dim Y' \geq \dim Y$ for the base Y' of any compression of f. Let p be a prime. We say that f is *p-incompressible* if for any variety Z over k and any dominant generically finite morphism $Z \to Y$ of degree prime to p, the morphism $X \times_Y Z \to Z$ is incompressible.

The degree of $Z \to Y$ in Definition 2.1 is assumed to be prime to p for the following reason. Let A be an abelian variety over k with $\dim A \geq 1$, and assume that the characteristic of k is not p. The base change of $[p]$ along $[p]$ is $A \times_{[p],A,[p]} A \cong A \times A[p]$ via $(x, z) \to (z, x - z)$, which is the pullback of $A[p] \to \operatorname{Spec} k$ and hence has a compression whose base is a point, whereas for any dominant generically finite morphism $Z \to A$ of degree prime to p, the scheme $A \times_{[p],A} Z$ is still irreducible, unlike $A \times A[p]$, since every irreducible component of $A \times_{[p],A} Z$ dominates Z by flatness of $[p]$, and comparing its degrees over A along both sides of the fiber product square shows that its degree over Z is divisible by $p^{2 \dim A}$, the degree of $A \times_{[p],A} Z$ over Z.

The following notion is taken from [BF24, §1.1]. There it is further assumed that $X \times_Y Z$ is reduced and that f' is finite étale. Since a compression only involves the generic points, both conditions hold after replacing Z with the preimage of a dense open subset of Y over which f is étale and Y' with a dense open subset (by [Gro67, Proposition 17.5.7], and by generic finiteness [Har77, Chapter II, Exercise 3.7] applied to the irreducible components of X' which dominate Y'), and the minimum below does not change.

Definition-Proposition 2.2. Let k be a field, p a prime, Y a variety over k, and $f : X \to Y$ a separated morphism of finite type which is étale over a dense open subset of Y. The *essential dimension at p* of f, denoted by $\operatorname{ed}_k(f; p)$, is the minimum of $\dim Y'$, where $Z \to Y$ runs through the dominant generically finite morphisms of degree prime to p from varieties Z over k, and Y' runs through the bases of the compressions of $X \times_Y Z \to Z$. Then $\operatorname{ed}_k(f; p) \leq \dim Y$, and f is p-incompressible if and only if $\operatorname{ed}_k(f; p) = \dim Y$.

Proof. Taking $Z = Y$ and the compression of f with $Y' = Y$ and $f' = f$, we obtain $\operatorname{ed}_k(f; p) \leq \dim Y$. For any dominant generically finite morphism $Z \to Y$, we have $\dim Z = \dim Y$. Thus $X \times_Y Z \to Z$ is incompressible if and only if $\dim Y' \geq \dim Y$ for the base Y' of any compression of $X \times_Y Z \to Z$. Hence f is p-incompressible if and only if $\operatorname{ed}_k(f; p) \geq \dim Y$, that is, $\operatorname{ed}_k(f; p) = \dim Y$. □

2.2. **The holomorphic Atiyah–Bott formula.** In this subsection the base field is $\mathbb{C}$. In Section 3, the coefficient complex is the derived pullback of the structure sheaf of a small diagonal along a product morphism $f \times \cdots \times f$, where f need not be flat, so it is a perfect complex rather than a sheaf and may have several non-zero cohomology sheaves. Thus we define Lefschetz numbers through hypercohomology and local terms through derived fibers.

Definition 2.3. A *perfect complex* on a scheme V is a complex of $\mathcal{O}_V$-modules which is locally quasi-isomorphic to a bounded complex of locally free sheaves of finite rank. On a smooth projective variety, the perfect complexes are exactly the dualizable objects of the derived category of quasi-coherent sheaves [BFN10, Proposition 3.6], see also [KP20, §2.1], and for any morphism g, the derived pullback $\mathbf{L}g^*$ sends perfect complexes to perfect complexes [Sta, Tag 09UA]. For a perfect complex $\mathcal{E}$ on V and a closed point $z \in V$, we call $\mathcal{E} \otimes^{\mathbf{L}} \kappa(z)$ the *derived fiber* of $\mathcal{E}$ at z. If $\mathcal{E}$ is a coherent sheaf, then for any integer j,

$$\mathcal{H}^{-j}(\mathcal{E} \otimes^{\mathbf{L}} \kappa(z)) = \operatorname{Tor}_j^{\mathcal{O}_{V,z}}(\mathcal{E}_z, \kappa(z)).$$

We denote by $H^j(V, \mathcal{E})$ the j-th hypercohomology of $\mathcal{E}$.

Definition 2.4. Let V be a smooth projective scheme, $u : V \to V$ an automorphism, $\mathcal{E}$ a perfect complex on V, and $\varphi : u^*\mathcal{E} \to \mathcal{E}$ an isomorphism in the derived category of V. For any integer j, let φ_j be the endomorphism of $H^j(V, \mathcal{E})$ obtained by composing the pullback u^* from $H^j(V, \mathcal{E})$ to $H^j(V, u^*\mathcal{E})$ with $H^j(\varphi)$. The *Lefschetz number* of (u, φ) is

$$L(u, \varphi) := \sum_j (-1)^j \operatorname{Tr}(\varphi_j),$$

where $\dim H^j(V, \mathcal{E}) < +\infty$ for any j and $H^j(V, \mathcal{E}) = 0$ for all but finitely many j, since V is projective and $\mathcal{E}$ is perfect. For a fixed point z of u, the isomorphism φ induces an automorphism $\varphi(z)$ of the derived fiber $\mathcal{E} \otimes^{\mathbf{L}} \kappa(z)$, and the *local term* of (u, φ) at z is

$$\operatorname{tr}(u, \varphi; z) := \sum_j (-1)^j \operatorname{Tr}\left(\varphi(z) \mid \mathcal{H}^j(\mathcal{E} \otimes^{\mathbf{L}} \kappa(z))\right).$$

The holomorphic Atiyah–Bott formula [AB67, AB68], in the form of [KP20, Theorem 3.1.2], holds for a smooth proper variety over an algebraically closed field with an endomorphism whose graph intersects the diagonal transversally, together with a lax equivariant structure on a dualizable object [KP20, Conventions 3.0.1]. Here the algebraically closed field is $\mathbb{C}$, the endomorphism is an automorphism u of a smooth projective variety, the lax equivariant structure is the isomorphism φ, and under transversality the fixed points of u are finite in number [KP20, Conventions 3.0.1]. The Lefschetz number in [KP20, Theorem 3.1.2] is the categorical trace of the endomorphism of the derived global sections of $\mathcal{E}$ induced by (u, φ), which equals the alternating sum of the traces on cohomology (cf. [KP20, Example 1.1.3(1)]), that is, $L(u, \varphi)$. The numerator in [KP20, Theorem 3.1.2] at a fixed point z is the trace on the derived stalk of $\mathcal{E}$ at z in the sense of [KP20, Proposition 3.2.3], an object of the derived category of $\mathbb{C}$-vector spaces, which is the derived fiber $\mathcal{E} \otimes^{\mathbf{L}} \kappa(z)$ in our notation. Transversality of the graph of u to the diagonal of $V \times V$ is equivalent to $\det(1 - \mathrm{d}u_z \mid T_z V) \neq 0$ for any fixed point z of u, as noted after the formula in [KP20, Theorem 3.1.2].

For the proof of Theorem 1.3, Section 3 constructs, for each $h \in A[p]$, an automorphism of finite order of a smooth projective scheme, and Theorem 2.5 is applied in Section 4 to the restrictions of the automorphism to the connected components which it maps to themselves.

Theorem 2.5 ([KP20, Theorem 3.1.2 and Proposition 3.2.3])**.** *Let V be a smooth projective variety, $u : V \to V$ an automorphism, $\mathcal{E}$ a perfect complex on V, and $\varphi : u^*\mathcal{E} \to \mathcal{E}$ an isomorphism in the derived category of V. Assume that the graph of u intersects the diagonal of $V \times V$ transversally, equivalently, $\det(1 - \mathrm{d}u_z \mid T_z V) \neq 0$ for any fixed point z of u. Then u has*

finitely many fixed points, and

$$L(u,\varphi) = \sum_{z\in V, u(z)=z} \frac{\operatorname{tr}(u,\varphi;z)}{\det(1-\mathrm{d}u_z \mid T_zV)}.$$

3. Fixed points and local terms on a fiber of the sum map

In this section, we work in Set-up 3.1, which is the situation of Theorem 1.3. This section carries out the local part of the argument for Theorem 1.3: we construct automorphisms of a closed subscheme of X^p with finitely many transverse fixed points, and compute all fixed points, denominators, and local terms that enter the holomorphic Atiyah–Bott formula (Theorem 2.5). The global computation is completed in Section 4.

Set-up 3.1. Let $p, d, n, A, G, X, T, q, r, f, \tau, s, c_p$ be as follows.

(1) p is a prime, and $d \geq 1$ and $0 \leq n \leq d$ are integers.
(2) A is an abelian variety of dimension d, and $G := A[p]$ acts on A by translations.
(3) X and T are smooth projective varieties of dimensions d and n respectively, with regular G-actions, and the G-action on T is faithful.
(4) $q : X \to A$ is a G-equivariant dominant generically finite morphism of degree r.
(5) $f : X \to T$ is a G-equivariant surjective morphism.
(6) G acts diagonally on X^p, and $\tau : X^p \to X^p$ is the automorphism
$$(x_0, \ldots, x_{p-1}) \to (x_1, \ldots, x_{p-1}, x_0).$$
(7) $s : X^p \to A$ is the morphism $(x_0, \ldots, x_{p-1}) \to \sum_{i=0}^{p-1} q(x_i)$, and $c_p := p(p-1)/2$.
(8) The group law of G is written additively.
(9) For $h \in G$ and a point x of A, X, T, or of a product of them, hx denotes the image of x under h, so that $hx = x + h$ for $x \in A$.

With ℓ factors instead of p, the diagonal action of $h \in G$ on X^ℓ would map the fiber of $(x_i) \to \sum_i q(x_i)$ over $c \in A$ to the fiber over $c + \ell h$, so the fibers are G-stable only if $p \mid \ell$.

Construction-Proposition 3.2 chooses the fiber of s on which we count the fixed points of the composites $h \circ \tau$, $h \in G$. Properties (1)–(3) below guarantee that the fiber is smooth, that q is étale at the coordinates of the fixed points, and, for $h \neq 0$, that f maps none of the coordinates into the fixed locus $\mathrm{Fix}_T(h)$ of h on T.

Construction-Proposition 3.2. Notation and conditions as in Set-up 3.1. There exists a closed point $a \in A$ with the following properties.

(1) The morphism s is smooth over a neighborhood of a, so that $s^{-1}(a)$ is a smooth projective scheme of pure dimension $(p-1)d$.
(2) For any $h \in G$ and any point $b \in A$ with $[p](b) = a + c_p h$, the morphism q is finite étale of degree r over a neighborhood of b.
(3) For any $0 \neq h \in G$ and any point $x \in X$ with $[p](q(x)) = a + c_p h$, we have $hf(x) \neq f(x)$.

Moreover, for any closed point $a \in A$, the fiber $s^{-1}(a)$ is stable under the diagonal action of G and under τ, and τ commutes with the diagonal action of G. We fix a closed point a with properties (1)–(3), and set $V := s^{-1}(a)$.

Proof. By [Har77, Chapter II, Exercise 3.7] and [Har77, Chapter III, Corollary 10.7], there exists a dense open subset $U \subset A$ such that q is finite étale of degree r over U.

For $h \in G$, let $\mathrm{Fix}_T(h) \subset T$ be the fixed locus of h on T. Let $0 \neq h \in G$. Since the G-action on T is faithful, $\mathrm{Fix}_T(h)$ is a proper closed subset of T. Since f is surjective, $f^{-1}(\mathrm{Fix}_T(h))$ is a proper closed subset of X. Since q is proper and X is irreducible of dimension d, $q(f^{-1}(\mathrm{Fix}_T(h)))$ is a closed subset of A and

$$\dim q(f^{-1}(\mathrm{Fix}_T(h))) \leq d - 1.$$

The morphism $[p] : A \to A$ is a finite étale isogeny of degree p^{2d} [Mum74, §6, Application 3 and §7, Theorem 4, Corollary 1]. Thus $[p](A\backslash U)$ and $[p](q(f^{-1}(\mathrm{Fix}_T(h))))$ for $0 \neq h \in G$ are proper closed subsets of A.

Since q is proper and dominant, q is surjective, hence s is surjective. Since X^p is smooth, by generic smoothness [Har77, Chapter III, Corollary 10.7], there exists a dense open subset $U' \subset A$ such that s is smooth over U'. Since $\dim X^p = pd$ and s is surjective, the fiber of s over any closed point of U' is a smooth projective scheme of pure dimension $(p-1)d$.

For a subset $W \subset A$ and $c \in A$, let $W - c := \{w - c \mid w \in W\}$. Since A is irreducible and G is finite, there exists a closed point $a \in U'$ which is not contained in the finite union of proper closed subsets

$$\bigcup_{h \in G} \big([p](A\backslash U) - c_p h\big) \cup \bigcup_{0 \neq h \in G} \big([p](q(f^{-1}(\mathrm{Fix}_T(h)))) - c_p h\big).$$

We show that a satisfies (1)–(3). Property (1) holds since $a \in U'$.

For (2), let $h \in G$ and let $b \in A$ be a point with $[p](b) = a + c_p h$. If $b \in A\backslash U$, then

$$a = [p](b) - c_p h \in [p](A\backslash U) - c_p h,$$

a contradiction. Thus $b \in U$, and q is finite étale of degree r over the neighborhood U of b.

For (3), let $0 \neq h \in G$ and let $x \in X$ be a point with $[p](q(x)) = a + c_p h$. If $hf(x) = f(x)$, then $x \in f^{-1}(\mathrm{Fix}_T(h))$, and

$$a = [p](q(x)) - c_p h \in [p](q(f^{-1}(\mathrm{Fix}_T(h)))) - c_p h,$$

a contradiction. Thus $hf(x) \neq f(x)$.

For the last statement, let $h \in G$ and $(x_0, \dots, x_{p-1}) \in X^p$. Since q is G-equivariant and $ph = 0$,

$$s(hx_0, \dots, hx_{p-1}) = \sum_{i=0}^{p-1} (q(x_i) + h) = s(x_0, \dots, x_{p-1}) + ph = s(x_0, \dots, x_{p-1}).$$

Moreover, $s \circ \tau = s$, and

$$\tau(hx_0, \dots, hx_{p-1}) = (hx_1, \dots, hx_{p-1}, hx_0)$$

is the image of $\tau(x_0, \dots, x_{p-1})$ under h. Thus any fiber of s is stable under the diagonal action of G and under τ, and τ commutes with the diagonal action of G. □

Construction-Proposition 3.3 brings T into the holomorphic Atiyah–Bott formula through a complex on V, the derived pullback under $f \times \cdots \times f$ of the structure sheaf of the small diagonal of T^p. By the associativity of derived pullback, the derived fiber of the complex at a closed point $z \in V$ is the derived fiber of the structure sheaf of the small diagonal at the image of z in T^p, so the properties (1)–(3) of the closed point a in Construction-Proposition 3.2 contain no condition on the smoothness of f.

Construction-Proposition 3.3. Notation and conditions as in Construction-Proposition 3.2. Let $\Gamma := G \times \mathbb{Z}/p\mathbb{Z}$ act on X^p and on T^p, where G acts diagonally and $1 \in \mathbb{Z}/p\mathbb{Z}$ acts by the cyclic permutation $(y_0, \dots, y_{p-1}) \to (y_1, \dots, y_{p-1}, y_0)$ of the factors. Thus $1 \in \mathbb{Z}/p\mathbb{Z}$ acts on X^p by τ, and V is Γ-stable. We write $h\tau$ for the element $(h, 1) \in \Gamma$. Let $\Delta \subset T^p$ be the small diagonal, let $f_V : V \to T^p$ be the restriction of $f \times \cdots \times f : X^p \to T^p$ to V, and set $\mathcal{E} := \mathbf{L}f_V^* \mathcal{O}_\Delta$. Then the following hold.

(1) The morphism f_V is Γ-equivariant, and $\mathcal{E}$ is a perfect complex on V.
(2) The Γ-equivariant structure of $\mathcal{O}_\Delta$ as a quotient of $\mathcal{O}_{T^p}$ induces isomorphisms

$$\varphi_\gamma : \gamma^* \mathcal{E} \to \mathcal{E}, \quad \gamma \in \Gamma,$$

such that

$$\varphi_{\gamma\delta} = \varphi_\delta \circ \delta^* \varphi_\gamma$$

for any $\gamma, \delta \in \Gamma$.

(3) For any integer j, the map ρ_j which sends $\gamma \in \Gamma$ to the composition of the pullback γ^* from $H^j(V, \mathcal{E})$ to $H^j(V, \gamma^*\mathcal{E})$ with $H^j(\varphi_\gamma)$ is a representation of Γ on $H^j(V, \mathcal{E})$.

Proof. By Construction-Proposition 3.2, V is stable under the diagonal action of G and under τ, hence V is Γ-stable. The small diagonal Δ is stable under Γ, so the ideal sheaf of Δ is Γ-stable, and $\mathcal{O}_\Delta$ inherits, as a quotient of $\mathcal{O}_{T^p}$, the Γ-equivariant structure of $\mathcal{O}_{T^p}$. For $\gamma \in \Gamma$, let ψ_γ be the isomorphism from $\gamma^*\mathcal{O}_\Delta$ to $\mathcal{O}_\Delta$ defining the action of γ in the Γ-equivariant structure of $\mathcal{O}_\Delta$. Then

$$\psi_{\gamma\delta} = \psi_\delta \circ \delta^*\psi_\gamma$$

for any $\gamma, \delta \in \Gamma$.

(1) Since f is G-equivariant, $f \times \cdots \times f$ commutes with the diagonal actions of G on X^p and on T^p, and with the cyclic permutations of the factors. Thus f_V is Γ-equivariant.

Since T is smooth, $\Delta \cong T$ is a smooth closed subvariety of codimension $(p-1)n$ of the smooth variety T^p. Let $w \in \Delta$ be a closed point, and let I be the ideal of Δ in $\mathcal{O}_{T^p,w}$. By [Har77, Chapter II, Theorem 8.17], I is generated by $(p-1)n = \operatorname{codim}(\Delta, T^p)$ elements, and I/I^2 is free of rank $(p-1)n$ over $\mathcal{O}_{\Delta,w}$. Since $\mathcal{O}_{T^p,w}$ is a regular local ring, it is Cohen–Macaulay [Mat86, Theorem 17.8], and

$$\operatorname{ht} I = \dim \mathcal{O}_{T^p,w} - \dim \mathcal{O}_{\Delta,w} = pn - n = (p-1)n$$

by [Mat86, Theorem 17.4(i)]. Thus any $(p-1)n$ generators of I form a regular sequence [Mat86, Theorem 17.4(iii)], and the Koszul complex on them is a finite free resolution of $\mathcal{O}_{\Delta,w}$ [Mat86, Theorem 16.5(i)]. The generators extend to sections of the ideal sheaf of Δ over an open neighborhood of w, and since the homology sheaves of the Koszul complex on the extended sections are coherent, the Koszul complex resolves $\mathcal{O}_\Delta$ on a possibly smaller open neighborhood of w. Moreover, $\mathcal{O}_\Delta$ is zero on $T^p \backslash \Delta$. Thus $\mathcal{O}_\Delta$ is a perfect complex on T^p, and $\mathcal{E} = \mathbf{L}f_V^*\mathcal{O}_\Delta$ is a perfect complex on V, since derived pullback preserves perfect complexes.

(2) Let $\gamma \in \Gamma$. Since γ is an automorphism, $\mathbf{L}\gamma^* = \gamma^*$. Since $f_V \circ \gamma = \gamma \circ f_V$ by (1), the functoriality of derived pullback induces an isomorphism of functors σ_γ from $\gamma^*\mathbf{L}f_V^*$ to $\mathbf{L}f_V^*\gamma^*$. We let φ_γ be the composition

$$\gamma^*\mathcal{E} = \gamma^*\mathbf{L}f_V^*\mathcal{O}_\Delta \xrightarrow{\sigma_\gamma} \mathbf{L}f_V^*\gamma^*\mathcal{O}_\Delta \xrightarrow{\mathbf{L}f_V^*\psi_\gamma} \mathbf{L}f_V^*\mathcal{O}_\Delta = \mathcal{E}.$$

Let $\gamma, \delta \in \Gamma$. Under the identifications $(\gamma\delta)^* = \delta^*\gamma^*$ on V and on T^p, the isomorphisms $\sigma_\bullet$ are compatible with composition (by the coherence of the isomorphisms $\mathbf{L}(g \circ g')^* \cong \mathbf{L}g'^*\mathbf{L}g^*$ of derived pullback), that is,

$$\sigma_{\gamma\delta} = \sigma_\delta(\gamma^*\mathcal{O}_\Delta) \circ \delta^*\sigma_\gamma$$

on $\mathcal{O}_\Delta$. By the naturality of σ_δ, we have

$$\sigma_\delta(\mathcal{O}_\Delta) \circ \delta^*\mathbf{L}f_V^*\psi_\gamma = \mathbf{L}f_V^*\delta^*\psi_\gamma \circ \sigma_\delta(\gamma^*\mathcal{O}_\Delta).$$

Thus

$$\begin{aligned}
\varphi_\delta \circ \delta^*\varphi_\gamma &= \mathbf{L}f_V^*\psi_\delta \circ \sigma_\delta(\mathcal{O}_\Delta) \circ \delta^*\mathbf{L}f_V^*\psi_\gamma \circ \delta^*\sigma_\gamma \\
&= \mathbf{L}f_V^*\psi_\delta \circ \mathbf{L}f_V^*\delta^*\psi_\gamma \circ \sigma_\delta(\gamma^*\mathcal{O}_\Delta) \circ \delta^*\sigma_\gamma \\
&= \mathbf{L}f_V^*(\psi_\delta \circ \delta^*\psi_\gamma) \circ \sigma_{\gamma\delta} = \mathbf{L}f_V^*\psi_{\gamma\delta} \circ \sigma_{\gamma\delta} = \varphi_{\gamma\delta}.
\end{aligned}$$

(3) Let j be an integer and $\gamma, \delta \in \Gamma$. For any morphism $\alpha : \mathcal{F} \to \mathcal{F}'$ of complexes on V, the pullback δ^* on hypercohomology satisfies $\delta^* \circ H^j(\alpha) = H^j(\delta^*\alpha) \circ \delta^*$. By (2),

$$\begin{aligned}
\rho_j(\gamma\delta) &= H^j(\varphi_\delta) \circ H^j(\delta^*\varphi_\gamma) \circ \delta^* \circ \gamma^* \\
&= H^j(\varphi_\delta) \circ \delta^* \circ H^j(\varphi_\gamma) \circ \gamma^* = \rho_j(\delta) \circ \rho_j(\gamma).
\end{aligned}$$

Taking $\gamma = \delta = (0,0)$, the identity element of Γ, in (2), we have $\varphi_{(0,0)} = \varphi_{(0,0)} \circ \varphi_{(0,0)}$, so $\varphi_{(0,0)}$ is the identity and $\rho_j((0,0))$ is the identity. Since Γ is commutative,

$$\rho_j(\gamma\delta) = \rho_j(\delta\gamma) = \rho_j(\gamma) \circ \rho_j(\delta).$$

Thus ρ_j is a representation of Γ on $H^j(V, \mathcal{E})$. □

The following lemma is used for the denominators in Lemma 3.5(2) and for the local terms in Lemma 3.6(2).

Lemma 3.4. *Let p be a prime, ζ a primitive p-th root of unity, and M a $\mathbb{C}$-vector space with $\dim M < +\infty$. Let*

$$P : M^{\oplus p} \to M^{\oplus p}, \quad (v_0, \dots, v_{p-1}) \to (v_1, \dots, v_{p-1}, v_0)$$

be the cyclic permutation of the summands, and let $S \subset M^{\oplus p}$ be the subspace of the vectors $(v_0, \dots, v_{p-1})$ with $\sum_{i=0}^{p-1} v_i = 0$. Then S is stable under P and P^{-1}, the eigenvalues of P and of P^{-1} on S are ζ^j, $1 \le j \le p-1$, each with multiplicity $\dim M$, and

$$\det(1 - P \mid S) = \det(1 - P^{-1} \mid S) = p^{\dim M}.$$

Proof. We write $M^{\oplus p} = \mathbb{C}^p \otimes_{\mathbb{C}} M$, so that $P = P_0 \otimes \mathrm{id}_M$, where P_0 is the cyclic permutation of the coordinates of $\mathbb{C}^p$. For $0 \le j \le p-1$, the vector $e_j := (1, \zeta^j, \zeta^{2j}, \dots, \zeta^{(p-1)j})$ satisfies $P_0 e_j = \zeta^j e_j$, and the sum of the coordinates of e_j is zero if $1 \le j \le p-1$. Since $e_1, \dots, e_{p-1}$ are eigenvectors of P_0 with distinct eigenvalues, they span the $(p-1)$-dimensional subspace $S_0 \subset \mathbb{C}^p$ of the vectors with coordinate sum zero, and $S = S_0 \otimes_{\mathbb{C}} M$. Thus S is stable under P and P^{-1}, the eigenvalues of P on S are ζ^j, $1 \le j \le p-1$, each with multiplicity $\dim M$, and the eigenvalues of P^{-1} on S are $\zeta^{-j} = \zeta^{p-j}$, $1 \le j \le p-1$, each with multiplicity $\dim M$, where $\{p - j \mid 1 \le j \le p-1\} = \{1, \dots, p-1\}$. Evaluating

$$\frac{t^p - 1}{t - 1} = \prod_{j=1}^{p-1} (t - \zeta^j)$$

at $t = 1$, we obtain $\prod_{j=1}^{p-1}(1 - \zeta^j) = p$. Thus

$$\det(1 - P \mid S) = \det(1 - P^{-1} \mid S) = \prod_{j=1}^{p-1} (1 - \zeta^j)^{\dim M} = p^{\dim M}.$$

□

Lemma 3.5. *Notation and conditions as in Construction-Proposition 3.3, and let $h \in G$. Then the following hold.*

1. *A point $z \in V$ is fixed by $h\tau$ if and only if $z = (x, (-h)x, (-2h)x, \dots, (-(p-1)h)x)$ for some $x \in X$ with $[p](q(x)) = a + c_p h$. In particular, $h\tau$ has exactly $p^{2d} r$ fixed points on V.*
2. *For any fixed point z of $h\tau$ on V, we have $\det\left(1 - \mathrm{d}(h\tau)_z \mid T_z V\right) = p^d$.*

Proof. (1) Let $z = (x_0, \dots, x_{p-1}) \in X^p$, where indices are taken modulo p. Then

$$h\tau(z) = (hx_1, \dots, hx_{p-1}, hx_0),$$

so z is fixed by $h\tau$ if and only if $x_i = hx_{i+1}$ for any i. If z is fixed by $h\tau$, then $x_{i+1} = (-h)x_i$ for any i, hence $x_i = (-ih)x_0$ for $0 \le i \le p-1$. Conversely, for any $x \in X$, the point $(x, (-h)x, \dots, (-(p-1)h)x)$ is fixed by $h\tau$, since $hx_0 = hx = (-(p-1)h)x = x_{p-1}$ by $ph = 0$. Since q is G-equivariant,

$$s(x, (-h)x, \dots, (-(p-1)h)x) = \sum_{i=0}^{p-1} (q(x) - ih) = [p](q(x)) - c_p h.$$

Thus $(x, (-h)x, \dots, (-(p-1)h)x) \in V$ if and only if $[p](q(x)) = a + c_p h$.

The map $x \to (x, (-h)x, \dots, (-(p-1)h)x)$ is injective, so the fixed points of $h\tau$ on V correspond bijectively to the points $x \in X$ with $[p](q(x)) = a + c_p h$. Since $[p]$ is finite étale of degree p^{2d} [Mum74, §6, Application 3 and §7, Theorem 4, Corollary 1], there are exactly p^{2d} points $b \in A$ with $[p](b) = a + c_p h$. By Construction-Proposition 3.2(2), for any such b, q is finite étale of

degree r over a neighborhood of b, so there are exactly r points $x \in X$ with $q(x) = b$. Thus $h\tau$ has exactly $p^{2d}r$ fixed points on V.

(2) By (1), $z = (x_0, \dots, x_{p-1})$ with $x_i = (-ih)x$ for some $x \in X$ with $[p](q(x)) = a + c_p h$. For $0 \le i \le p-1$, we have $q(x_i) = q(x) - ih$, so

$$[p](q(x_i)) = [p](q(x)) - pih = a + c_p h.$$

By Construction-Proposition 3.2(2), q is étale at x_i, so $\mathrm{d}q_{x_i} : T_{x_i}X \to T_{q(x_i)}A$ is an isomorphism. For $c \in A$, let $t_c : A \to A$ be the translation $b \to b + c$. For $0 \le i \le p-1$, let

$$\iota_i := \mathrm{d}(t_{-q(x_i)})_{q(x_i)} \circ \mathrm{d}q_{x_i} : T_{x_i}X \to T_0 A,$$

which is an isomorphism, and let $\iota_p := \iota_0$. Since s is the composition of $q \times \cdots \times q$ with the addition $A^p \to A$, and the differential of the addition at $(0, \dots, 0)$ is the summation, under the isomorphism $\iota_0 \oplus \cdots \oplus \iota_{p-1}$ the differential $\mathrm{d}s_z$ becomes the summation $(T_0A)^{\oplus p} \to T_0A$, followed by $\mathrm{d}(t_a)_0$. Since s is smooth at z by Construction-Proposition 3.2(1), $T_zV = \ker(\mathrm{d}s_z)$. Thus $\iota_0 \oplus \cdots \oplus \iota_{p-1}$ maps T_zV isomorphically onto

$$S := \Big\{(v_0, \dots, v_{p-1}) \in (T_0A)^{\oplus p} \;\Big|\; \sum_{i=0}^{p-1} v_i = 0\Big\}.$$

Next we compute $\mathrm{d}(h\tau)_z$. Since $h\tau(z') = (hx'_1, \dots, hx'_{p-1}, hx'_0)$ for $z' = (x'_0, \dots, x'_{p-1}) \in X^p$ and $hx_{i+1} = x_i$, we have

$$\mathrm{d}(h\tau)_z(w_0, \dots, w_{p-1}) = (\mathrm{d}h_{x_1}w_1, \dots, \mathrm{d}h_{x_{p-1}}w_{p-1}, \mathrm{d}h_{x_0}w_0)$$

for $w_i \in T_{x_i}X$, where $\mathrm{d}h_{x_{i+1}} : T_{x_{i+1}}X \to T_{x_i}X$. Since $q \circ h = t_h \circ q$ and $q(x_i) = q(x_{i+1}) + h$,

$$\iota_i \circ \mathrm{d}h_{x_{i+1}} = \mathrm{d}(t_{-q(x_i)})_{q(x_i)} \circ \mathrm{d}(t_h)_{q(x_{i+1})} \circ \mathrm{d}q_{x_{i+1}} = \mathrm{d}(t_{-q(x_{i+1})})_{q(x_{i+1})} \circ \mathrm{d}q_{x_{i+1}} = \iota_{i+1}.$$

Thus, under $\iota_0 \oplus \cdots \oplus \iota_{p-1}$, $\mathrm{d}(h\tau)_z$ becomes the restriction to S of the cyclic permutation P of Lemma 3.4 with $M = T_0A$.

By Lemma 3.4 with $M = T_0A$, we obtain

$$\det\big(1 - \mathrm{d}(h\tau)_z \mid T_zV\big) = \det(1 - P \mid S) = p^d.$$

□

Lemma 3.6. *Notation and conditions as in Construction-Proposition 3.3. Then the following hold.*

(1) For any $0 \ne h \in G$ and any fixed point z of $h\tau$ on V, we have $\mathcal{E} \otimes^{\mathbf{L}} \kappa(z) = 0$. In particular, $\mathrm{tr}(h\tau, \varphi_{h\tau}; z) = 0$.

(2) For any fixed point z of τ on V, we have $\mathrm{tr}(\tau, \varphi_\tau; z) = p^n$.

Proof. (1) By Lemma 3.5(1), $z = (x, (-h)x, \dots, (-(p-1)h)x)$ for some $x \in X$ with $[p](q(x)) = a + c_p h$. Since f is G-equivariant,

$$f_V(z) = (f(x), (-h)f(x), \dots, (-(p-1)h)f(x)).$$

Since $hy = y$ if and only if $y = (-h)y$ for $y \in T$, we have $\mathrm{Fix}_T(-h) = \mathrm{Fix}_T(h)$. By Construction-Proposition 3.2(3), $hf(x) \ne f(x)$, so $f(x) \notin \mathrm{Fix}_T(-h)$, that is, $(-h)f(x) \ne f(x)$. Thus the first two coordinates of $f_V(z)$ are different, and $f_V(z) \notin \Delta$. Since Δ is closed, $T^p \backslash \Delta$ is an open neighborhood of $f_V(z)$, and $\mathcal{O}_\Delta|_{T^p\backslash\Delta} = 0$. Since derived pullback commutes with restriction to open subsets,

$$\mathcal{E}|_{f_V^{-1}(T^p\backslash\Delta)} = \mathbf{L}(f_V|_{f_V^{-1}(T^p\backslash\Delta)})^*(\mathcal{O}_\Delta|_{T^p\backslash\Delta}) = 0,$$

and $f_V^{-1}(T^p \backslash \Delta)$ is an open neighborhood of z. Thus $\mathcal{E} \otimes^{\mathbf{L}} \kappa(z) = 0$. In particular, $\mathcal{H}^j(\mathcal{E} \otimes^{\mathbf{L}} \kappa(z)) = 0$ for any j, and $\mathrm{tr}(h\tau, \varphi_{h\tau}; z) = 0$.

(2) By Lemma 3.5(1) with $h = 0$, $z = (x, \dots, x)$ for some $x \in X$ with $[p](q(x)) = a$. Let $y := f(x)$ and $w := f_V(z) = (y, \dots, y) \in \Delta$. The point w is fixed by τ. Let ψ_τ be the isomorphism from $\tau^*\mathcal{O}_\Delta$ to $\mathcal{O}_\Delta$ defining the action of τ in the Γ-equivariant structure of $\mathcal{O}_\Delta$ as a quotient of

$\mathcal{O}_{T^p}$, and let $\psi_\tau(w)$ be the automorphism of $\mathcal{O}_\Delta \otimes^{\mathbf{L}} \kappa(w)$ induced by ψ_τ. Let $i_z : \operatorname{Spec}\kappa(z) \to V$ and $i_w : \operatorname{Spec}\kappa(w) \to T^p$ be the inclusions. Then $f_V \circ i_z = i_w$, and the associativity of derived pullback induces an isomorphism

$$\mathcal{E} \otimes^{\mathbf{L}} \kappa(z) = \mathbf{L}i_z^*\mathbf{L}f_V^*\mathcal{O}_\Delta \cong \mathbf{L}i_w^*\mathcal{O}_\Delta = \mathcal{O}_\Delta \otimes^{\mathbf{L}} \kappa(w),$$

which is compatible with the automorphisms $\varphi_\tau(z)$ and $\psi_\tau(w)$, since φ_τ is induced by ψ_τ through the functoriality of derived pullback (Construction-Proposition 3.3(2)). Thus

$$\operatorname{tr}(\tau, \varphi_\tau; z) = \sum_j (-1)^j \operatorname{Tr}\left(\psi_\tau(w) \mid \mathcal{H}^j(\mathcal{O}_\Delta \otimes^{\mathbf{L}} \kappa(w))\right).$$

Let $R := \mathcal{O}_{T^p,w}$ with maximal ideal $\mathfrak{m}_w$, let $I \subset R$ be the ideal of Δ, and let $N := I/\mathfrak{m}_w I$. Since τ fixes w and Δ is τ-stable, the pullback of functions $g \to g \circ \tau$ is an automorphism of the local ring R which preserves I, and $\psi_\tau(w)$ is induced by it. We denote the automorphism $g \to g \circ \tau$ of R and the induced linear automorphism of N by τ, so $\tau^p = 1$. By [Har77, Chapter II, Theorem 8.17], I/I^2 is free of rank $(p-1)n$ over R/I, so $\dim N = \dim(I/I^2) \otimes_{R/I} \kappa(w) = (p-1)n$. By [Mat86, Theorem 2.3], any minimal set of generators of I consists of the lifts of a basis of N, and has exactly $(p-1)n$ elements. Since R is regular and $\operatorname{ht} I = (p-1)n$, any minimal set of generators of I is a regular sequence [Mat86, Theorems 17.8 and 17.4(i)(iii)].

We choose a minimal set of generators of I spanning a τ-stable subspace. Take a basis of N and lifts of the basis vectors to I. The images of the lifts under τ^k, $0 \le k \le p-1$, span a τ-stable $\mathbb{C}$-subspace $W_1 \subset I$ with $\dim W_1 < +\infty$, and the natural map $W_1 \to N$ is τ-equivariant and surjective. By [Ser77, §1.3, Theorem 1], applied to the group of order p generated by τ, the kernel of $W_1 \to N$ has a τ-stable complement $W_2 \subset W_1$, and $W_2 \to N$ is a τ-equivariant isomorphism. Let $\xi_1, \dots, \xi_{(p-1)n}$ be a basis of W_2. By [Mat86, Theorem 2.3], $\xi_1, \dots, \xi_{(p-1)n}$ is a minimal set of generators of I, hence a regular sequence.

Let $K_\bullet$ be the Koszul complex of $\xi_1, \dots, \xi_{(p-1)n}$, that is, $K_j = R \otimes_{\mathbb{C}} \wedge^j W_2$ with the differential induced by the inclusion $W_2 \subset R$. Since W_2 is τ-stable, τ acts on $K_\bullet$ compatibly with the action of τ on R and on R/I. By [Mat86, Theorem 16.5(i)], $K_\bullet$ is a free resolution of $R/I = \mathcal{O}_{\Delta,w}$. The differentials of $K_\bullet$ have entries in $\{0, \pm\xi_1, \dots, \pm\xi_{(p-1)n}\} \subset \mathfrak{m}_w$, so the differentials of $K_\bullet \otimes_R \kappa(w)$ are zero. Thus, for any $j \ge 0$, we have τ-equivariant isomorphisms

$$\mathcal{H}^{-j}(\mathcal{O}_\Delta \otimes^{\mathbf{L}} \kappa(w)) = \operatorname{Tor}_j^R(\mathcal{O}_{\Delta,w}, \kappa(w)) \cong \wedge^j W_2 \cong \wedge^j N,$$

and the cohomology sheaves of $\mathcal{O}_\Delta \otimes^{\mathbf{L}} \kappa(w)$ vanish in degrees $j > 0$ and $j < -(p-1)n$. Moreover, τ acts trivially on $\operatorname{Tor}_0^R(\mathcal{O}_{\Delta,w}, \kappa(w))$, which is $\kappa(w)$, since ψ_τ is induced by the equivariant structure of $\mathcal{O}_{T^p}$ and τ fixes w. Thus $\psi_\tau(w)$ acts on $\wedge^j N$ by $\wedge^j \tau$.

Next we compute the action of τ on N. Let $\mathfrak{m}_y$ be the maximal ideal of $\mathcal{O}_{T,y}$. The cotangent space $\mathfrak{m}_w/\mathfrak{m}_w^2$ of T^p at w is $(\mathfrak{m}_y/\mathfrak{m}_y^2)^{\oplus p}$, where the i-th summand is the pullback under the i-th projection $T^p \to T$. Since the composition of the i-th projection with τ is the $(i+1)$-th projection, indices modulo p, τ maps $(u_0, \dots, u_{p-1})$ to $(u_{p-1}, u_0, \dots, u_{p-2})$. The composition of the diagonal embedding $T \to T^p$ with any projection $T^p \to T$ is the identity, so the restriction map from $\mathfrak{m}_w/\mathfrak{m}_w^2$ to the cotangent space $\mathfrak{m}_y/\mathfrak{m}_y^2$ of $\Delta \cong T$ at w is

$$(u_0, \dots, u_{p-1}) \to \sum_{i=0}^{p-1} u_i.$$

The restriction map is surjective, and the kernel is $(I + \mathfrak{m}_w^2)/\mathfrak{m}_w^2$, so

$$\dim(I + \mathfrak{m}_w^2)/\mathfrak{m}_w^2 = pn - n = (p-1)n = \dim N.$$

Since the natural map $N \to (I + \mathfrak{m}_w^2)/\mathfrak{m}_w^2$ is surjective and τ-equivariant, it is a τ-equivariant isomorphism. Thus

$$N \cong \left\{(u_0, \dots, u_{p-1}) \in (\mathfrak{m}_y/\mathfrak{m}_y^2)^{\oplus p} \;\middle|\; \sum_{i=0}^{p-1} u_i = 0\right\}.$$

Thus τ acts on N as the restriction to S of the inverse P^{-1} of the cyclic permutation P in Lemma 3.4 with $M = \mathfrak{m}_y/\mathfrak{m}_y^2$, and by Lemma 3.4, for a primitive p-th root of unity ζ, the eigenvalues of τ on N are ζ^j, $1 \le j \le p-1$, each with multiplicity n, and $\det(1-\tau \mid N) = p^n$.

Let $\lambda_1, \dots, \lambda_{(p-1)n}$ be the eigenvalues of τ on N, counted with multiplicity. The eigenvalues of $\wedge^j\tau$ on $\wedge^j N$ are the products $\lambda_{k_1} \cdots \lambda_{k_j}$ for $k_1 < \cdots < k_j$, so

$$\sum_{j=0}^{(p-1)n} (-1)^j \operatorname{Tr}(\wedge^j\tau \mid \wedge^j N) = \prod_{k=1}^{(p-1)n} (1-\lambda_k) = \det(1-\tau \mid N).$$

Since the $(-j)$-th cohomology sheaf of $\mathcal{O}_\Delta \otimes^{\mathbf{L}} \kappa(w)$ is isomorphic to $\wedge^j N$ and $(-1)^{-j} = (-1)^j$, we obtain

$$\operatorname{tr}(\tau, \varphi_\tau; z) = \det(1-\tau \mid N) = p^n.$$

□

4. Proofs of the main theorems

In this section, we prove Theorem 1.3, and then deduce Theorem 1.2 from it.

4.1. **Proof of Theorem 1.3.**

Proof of Theorem 1.3. The data $p, d, n, A, G, X, T, q, r, f$ of Theorem 1.3 satisfy conditions (1)–(5) of Set-up 3.1. Let τ, s, and c_p be the objects defined in Set-up 3.1(6)(7), and let a, V, Γ, $\mathcal{E}$, φ_γ, and ρ_j be the objects of Construction-Propositions 3.2 and 3.3. For $h \in G$ we write $\rho_j(h) := \rho_j((h,0))$ and $\rho_j(\tau) := \rho_j((0,1))$.

Fix $h \in G$. By Definition 2.4 and Construction-Proposition 3.3(3),

$$L(h\tau, \varphi_{h\tau}) = \sum_j (-1)^j \operatorname{Tr}\big(\rho_j(h\tau) \mid H^j(V, \mathcal{E})\big).$$

The scheme V may be disconnected. Let $V_1, \dots, V_m$ be the connected components of V, so that

$$H^j(V, \mathcal{E}) = \bigoplus_{i=1}^m H^j(V_i, \mathcal{E}|_{V_i}).$$

The automorphism $h\tau$ of V permutes $V_1, \dots, V_m$, and $\rho_j(h\tau)$ maps $H^j(V_i, \mathcal{E}|_{V_i})$ into

$$H^j\big((h\tau)^{-1}(V_i), \mathcal{E}|_{(h\tau)^{-1}(V_i)}\big).$$

Thus only the components V_i with $h\tau(V_i) = V_i$ contribute to the trace of $\rho_j(h\tau)$. Hence

$$L(h\tau, \varphi_{h\tau}) = \sum_{i : h\tau(V_i) = V_i} L\big(h\tau|_{V_i}, \varphi_{h\tau}|_{V_i}\big). \tag{4.1}$$

We check the hypotheses of Theorem 2.5 for each V_i with $h\tau(V_i) = V_i$.

- V_i is a smooth projective variety: V is a smooth projective scheme by Construction-Proposition 3.2(1), and V_i is smooth and connected, hence integral.
- $u := h\tau|_{V_i} : V_i \to V_i$ is an automorphism, since V is Γ-stable and $h\tau(V_i) = V_i$.
- $\mathcal{E}|_{V_i}$ is a perfect complex, and

  $$\varphi_{h\tau}|_{V_i} : u^*(\mathcal{E}|_{V_i}) \to \mathcal{E}|_{V_i}$$

  is an isomorphism in the derived category of V_i (Construction-Proposition 3.3(1)(2)).
- For any fixed point z of u, we have $T_z V_i = T_z V$ and $\det(1 - \mathrm{d}u_z \mid T_z V_i) = p^d \neq 0$ (Lemma 3.5(2)).

The local term of $(u, \varphi_{h\tau}|_{V_i})$ at z is equal to $\mathrm{tr}(h\tau, \varphi_{h\tau}; z)$, since the derived fiber of $\mathcal{E}|_{V_i}$ at z is equal to the derived fiber of $\mathcal{E}$ at z, and both local terms are computed on it. Since any fixed point of $h\tau$ is contained in a component V_i with $h\tau(V_i) = V_i$, Theorem 2.5, (4.1), and Lemma 3.5(2) give

$$L(h\tau, \varphi_{h\tau}) = \frac{1}{p^d} \sum_{z \in V, h\tau(z)=z} \mathrm{tr}(h\tau, \varphi_{h\tau}; z). \tag{4.2}$$

If $h \neq 0$, then any summand of (4.2) is zero by Lemma 3.6(1). If $h = 0$, then τ has exactly $p^{2d}r$ fixed points on V by Lemma 3.5(1), and each local term is p^n by Lemma 3.6(2). Therefore

$$L(h\tau, \varphi_{h\tau}) = \begin{cases} 0 & \text{if } h \neq 0, \\ p^{2d}r \cdot p^n/p^d = p^{d+n}r & \text{if } h = 0. \end{cases} \tag{4.3}$$

Since Γ is abelian and ρ_j is a representation, $\rho_j(h\tau) = \rho_j(\tau)\rho_j(h)$ for any $h \in G$. Let

$$P_j := \frac{1}{|G|} \sum_{h \in G} \rho_j(h).$$

Then P_j is the projection of $H^j(V, \mathcal{E})$ onto the subspace $H^j(V, \mathcal{E})^G$ of G-invariant vectors, and P_j commutes with $\rho_j(\tau)$. Thus $\rho_j(\tau)$ preserves both $H^j(V, \mathcal{E})^G$ and $\operatorname{Ker} P_j$, and

$$\mathrm{Tr}\left(\rho_j(\tau) \mid H^j(V, \mathcal{E})^G\right) = \mathrm{Tr}\left(\rho_j(\tau)P_j \mid H^j(V, \mathcal{E})\right) = \frac{1}{|G|} \sum_{h \in G} \mathrm{Tr}\left(\rho_j(h\tau) \mid H^j(V, \mathcal{E})\right).$$

Since $|G| = p^{2d}$ [Mum74, §6, Application 3], taking the alternating sum over j and using (4.3), we obtain

$$\sum_j (-1)^j \mathrm{Tr}\left(\rho_j(\tau) \mid H^j(V, \mathcal{E})^G\right) = \frac{1}{p^{2d}} \sum_{h \in G} L(h\tau, \varphi_{h\tau}) = \frac{p^{d+n}r}{p^{2d}} = \frac{r}{p^{d-n}}. \tag{4.4}$$

Since $\tau^p = 1$ in Γ, we have $\rho_j(\tau)^p = 1$. Thus $\rho_j(\tau)$ is diagonalizable on $H^j(V, \mathcal{E})^G$, and the eigenvalues of $\rho_j(\tau)$ on $H^j(V, \mathcal{E})^G$ are p-th roots of unity. Hence the left-hand side of (4.4) is a linear combination of p-th roots of unity with integer coefficients, and so it is an algebraic integer [AM69, Proposition 5.1 and Corollary 5.3]. The right-hand side of (4.4) is a rational number. A rational number which is an algebraic integer is an integer [AM69, Example 5.0]. Thus $r/p^{d-n} \in \mathbb{Z}$, that is, p^{d-n} divides r. □

4.2. **Proof of Theorem 1.2.** In this subsection, we deduce Theorem 1.2 from Theorem 1.3.

The following lemma reduces Theorem 1.2 to the case when the base field is $\mathbb{C}$. The only point is that an integral variety over a subfield of k which is finitely generated over $\mathbb{Q}$ may become reducible after base change along a field embedding of the subfield into $\mathbb{C}$, and the degrees of the irreducible components over A' add up to a number prime to p, so one component has degree prime to p. Only the cover $[p]$ is used, never the group scheme $A[p]$ over k, so the case when $A[p]$ is not split needs no separate treatment.

Lemma 4.1. *Let k be a field of characteristic zero, A an abelian variety of dimension d over k, and p a prime. Assume that $[p] : A \to A$ is not p-incompressible. Then there exists an abelian variety A' of dimension d over $\mathbb{C}$ such that $[p] : A' \to A'$ is not p-incompressible.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we define all data over a subfield $k_0 \subset k$ which is finitely generated over $\mathbb{Q}$.

Since $[p]$ is not p-incompressible, there exist a variety Z over k, a dominant generically finite morphism $g : Z \to A$ of degree r with $p \nmid r$, and a compression of

$$W := A \times_{[p], A, g} Z \to Z,$$

consisting of a variety Z' over k, a dominant rational map $Z \dashrightarrow Z'$, and a morphism $W' \to Z'$ as in Definition 2.1, such that $\dim Z' < d$. We fix an isomorphism over the generic point of Z which exhibits $W \to Z$ as the pullback of $W' \to Z'$. Since $W' \to Z'$ is étale over a dense open subset of Z' (Definition 2.1), any irreducible component of W' which dominates Z', with the reduced structure, is finite over a dense open subset of Z' [Har77, Chapter II, Exercise 3.7]. Removing from Z' the closure of the image of the irreducible components of W' which do not dominate Z', we may assume, after shrinking Z' to a dense affine open subset, that $W' \to Z'$ is finite étale. Fix a dense open subset $Z^\circ \subset Z$ such that the rational map $Z \dashrightarrow Z'$ is a morphism $\psi : Z^\circ \to Z'$ on Z°, and such that the isomorphism fixed above extends to an isomorphism $W|_{Z^\circ} \cong W' \times_{Z'} Z^\circ$ over Z° [Gro66, Corollary 8.8.2.5].

The field k is the union of the subfields of k which are finitely generated over $\mathbb{Q}$. By [Gro66, Theorem 8.8.2], there exist a subfield $k_0 \subset k$ finitely generated over $\mathbb{Q}$ and the following data over k_0, whose base changes along $k_0 \to k$ are the data above: a scheme A_0 with a unit section, a multiplication morphism, and an inverse morphism, which satisfy the group axioms; a morphism $g_0 : Z_0 \to A_0$; a morphism $W_0' \to Z_0'$; an open subscheme $Z_0^\circ \subset Z_0$; a morphism $\psi_0 : Z_0^\circ \to Z_0'$; and an isomorphism

$$W_0|_{Z_0^\circ} \cong W_0' \times_{Z_0'} Z_0^\circ$$

over Z_0°, where $W_0 := A_0 \times_{[p],A_0,g_0} Z_0$. Possibly enlarging k_0, the morphisms $A_0 \to \operatorname{Spec} k_0$ and $W_0' \to Z_0'$ are proper and finite respectively by [Gro66, Theorem 8.10.5], and $A_0 \to \operatorname{Spec} k_0$ and $W_0' \to Z_0'$ are smooth and étale respectively by [Gro67, Proposition 17.7.8].

For any field extension K/k_0 and any scheme Y of finite type over k_0, the morphism $Y \otimes_{k_0} K \to Y$ is surjective and the map from $\mathcal{O}_Y$ to the pushforward of $\mathcal{O}_{Y \otimes_{k_0} K}$ is injective. Hence Z_0 and Z_0' are integral, since $Z_0 \otimes_{k_0} k = Z$ and $Z_0' \otimes_{k_0} k = Z'$ are integral, and they are separated over k_0, since separatedness descends along $\operatorname{Spec} k \to \operatorname{Spec} k_0$ [Gro65, Proposition 2.7.1]. Similarly, A_0 is geometrically integral, since $A_0 \otimes_{k_0} \bar{k}$ is integral for an algebraic closure $\bar{k}$ of k which contains an algebraic closure of k_0. Thus A_0 is an abelian variety over k_0. Dimensions are invariant under field extensions, so $\dim A_0 = d$, $\dim Z_0 = d$, and $\dim Z_0' = \dim Z' < d$. The morphism g_0 is dominant since g is dominant, and $\dim Z_0 = \dim A_0$, so g_0 is generically finite. By [Har77, Chapter II, Exercise 3.7] and [Gro65, Théorème 6.9.1], there exists a dense open subset $U_0 \subset A_0$ such that g_0 is finite and flat over U_0. The degree of g_0 is the rank of $(g_0)_* \mathcal{O}_{Z_0}$ over U_0, and the rank is invariant under base change. Hence $\deg g_0 = \deg g = r$.

Step 2. In this step we pass from k_0 to $\mathbb{C}$.

The field k_0 is a finite extension of a purely transcendental extension $\mathbb{Q}(t_1, \dots, t_m)$ of $\mathbb{Q}$. Since $\mathbb{C}$ is algebraically closed and the transcendence degree of $\mathbb{C}$ over $\mathbb{Q}$ is uncountable, there exists a field embedding $k_0 \to \mathbb{C}$, which we fix. Let

$$A' := A_0 \otimes_{k_0} \mathbb{C}.$$

Then A' is an abelian variety of dimension d over $\mathbb{C}$, and $[p] : A' \to A'$ is the base change of $[p] : A_0 \to A_0$.

Step 3. In this step we find an irreducible component Z_1 of $Z_0 \otimes_{k_0} \mathbb{C}$ such that $Z_1 \to A'$ is dominant generically finite of degree prime to p.

By **Step 1**, the morphism

$$g_0 \otimes_{k_0} \mathbb{C} : Z_0 \otimes_{k_0} \mathbb{C} \to A'$$

is finite and flat of degree r over $U_0 \otimes_{k_0} \mathbb{C}$. The scheme $Z_0 \otimes_{k_0} \mathbb{C}$ is reduced [Mat86, Theorem 26.3], since k_0 has characteristic zero. The morphism $Z_0 \otimes_{k_0} \mathbb{C} \to Z_0$ is flat. By the going-down theorem for flat morphisms [Mat86, Theorem 9.5], the generic point of any irreducible component of $Z_0 \otimes_{k_0} \mathbb{C}$ maps to the generic point of Z_0, hence is contained in the preimage of $U_0 \otimes_{k_0} \mathbb{C}$. Since $g_0 \otimes_{k_0} \mathbb{C}$ is flat over $U_0 \otimes_{k_0} \mathbb{C}$, the going-down theorem [Mat86, Theorem 9.5] again implies that the generic point of any irreducible component of $Z_0 \otimes_{k_0} \mathbb{C}$ maps to the generic point of A'. Thus any irreducible component of $Z_0 \otimes_{k_0} \mathbb{C}$ dominates A'.

Let $Z_{(1)}, \dots, Z_{(l)}$ be the irreducible components of $Z_0 \otimes_{k_0} \mathbb{C}$ with the reduced structures. Over the dense open subset of $U_0 \otimes_{k_0} \mathbb{C}$ obtained by removing the images of $Z_{(i)} \cap Z_{(i')}$ for $i \neq i'$, which are closed of dimension less than d, the scheme $Z_0 \otimes_{k_0} \mathbb{C}$ is the disjoint union of $Z_{(1)}, \dots, Z_{(l)}$. Hence

$$\sum_{i=1}^{l} \deg\left(Z_{(i)} \to A'\right) = r.$$

Since $p \nmid r$, there exists i such that $p \nmid \deg(Z_{(i)} \to A')$. Let $Z_1 := Z_{(i)}$ and $r_1 := \deg(Z_1 \to A')$. Then Z_1 is a variety over $\mathbb{C}$, and $Z_1 \to A'$ is a dominant generically finite morphism of degree r_1 with $p \nmid r_1$.

Step 4. In this step we restrict the compression to Z_1.

Let $Z_1^\circ := Z_1 \cap (Z_0^\circ \otimes_{k_0} \mathbb{C})$. Since the generic point of Z_1 maps to the generic point of Z_0 (**Step 3**), Z_1° is a dense open subset of Z_1. Let Z_1' be the closure of the image of Z_1° under $\psi_0 \otimes_{k_0} \mathbb{C}$ in $Z_0' \otimes_{k_0} \mathbb{C}$, with the reduced structure. Then Z_1' is a variety over $\mathbb{C}$, the rational map $Z_1 \dashrightarrow Z_1'$ is dominant, and

$$\dim Z_1' \leq \dim(Z_0' \otimes_{k_0} \mathbb{C}) = \dim Z_0' < d.$$

Let

$$W_1' := (W_0' \otimes_{k_0} \mathbb{C}) \times_{Z_0' \otimes_{k_0} \mathbb{C}} Z_1'.$$

Then $W_1' \to Z_1'$ is finite étale, as a base change of $W_0' \to Z_0'$.

Since fiber products commute with base change,

$$W_0 \otimes_{k_0} \mathbb{C} = A' \times_{[p],A'} (Z_0 \otimes_{k_0} \mathbb{C}).$$

Since Z_1° is reduced, $\psi_0 \otimes_{k_0} \mathbb{C}$ maps Z_1° into Z_1'. Base-changing the isomorphism over Z_0° of **Step 1** along $\mathbb{C}$ and restricting it to Z_1°, we obtain an isomorphism over Z_1°

$$\left(A' \times_{[p],A'} Z_1\right)|_{Z_1^\circ} \cong (W_0' \otimes_{k_0} \mathbb{C}) \times_{Z_0' \otimes_{k_0} \mathbb{C}} Z_1^\circ = W_1' \times_{Z_1'} Z_1^\circ.$$

Thus Z_1', $Z_1 \dashrightarrow Z_1'$, and $W_1' \to Z_1'$ form a compression of $A' \times_{[p],A'} Z_1 \to Z_1$ with $\dim Z_1' < d$ and $\dim Z_1 = d$. Hence $A' \times_{[p],A'} Z_1 \to Z_1$ is not incompressible. Since $Z_1 \to A'$ is dominant generically finite of degree r_1 with $p \nmid r_1$, $[p] : A' \to A'$ is not p-incompressible. □

In the following lemma, the base field is $\mathbb{C}$. The lemma turns the failure of p-incompressibility of $[p]$ into the equivariant smooth projective data of Theorem 1.3. For a dominant generically finite morphism $Z \to A$ of degree prime to p, a compression of $A \times_{[p],A} Z \to Z$ need not carry an action of G, whereas Theorem 1.3 requires equivariant data and a faithful action (cf. [BF24, Lemma 2] for torsors).

Lemma 4.2. *Let A be an abelian variety of dimension $d \geq 1$, p a prime, and $G := A[p]$ acting on A by translations. Assume that $[p] : A \to A$ is not p-incompressible. Then there exist smooth projective varieties X and T with regular G-actions, a G-equivariant dominant generically finite morphism $q : X \to A$ of degree prime to p, and a G-equivariant surjective morphism $f : X \to T$, such that $\dim X = d$, $\dim T < d$, and the G-action on T is faithful.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix a compression and reduce to the case when Z is projective.

Since $[p]$ is not p-incompressible, there exist a variety Z, a dominant generically finite morphism $g : Z \to A$ of degree r with $p \nmid r$, and a compression of

$$W := A \times_{[p],A,g} Z \to Z,$$

consisting of a variety Z', a dominant rational map $Z \dashrightarrow Z'$, and a morphism $W' \to Z'$ as in Definition 2.1, such that $\dim Z' < d$. We fix an isomorphism over the generic point of Z which exhibits $W \to Z$ as the pullback of $W' \to Z'$.

A compression of $W \to Z$ depends only on W over the generic point of Z. Let $U \subset Z$ be a dense affine open subset, fix a closed embedding $U \subset \mathbb{A}^N \subset \mathbb{P}^N$, and let $\bar{Z} \subset \mathbb{P}^N \times A$ be the closure of the image of $U \to \mathbb{P}^N \times A$, $z \to (z, g(z))$. Then $\bar{Z}$ is a projective variety which contains U as a dense open subset, and the second projection $\bar{Z} \to A$ extends $g|_U$. Thus $\bar{Z} \to A$ is dominant generically finite of degree r, and the compression of $W \to Z$ is a compression of $A \times_{[p],A} \bar{Z} \to \bar{Z}$ with base Z'. Possibly replacing Z with $\bar{Z}$, we may assume that Z is projective. We do not need to resolve the singularities of Z.

Let $\pi : W \to Z$ be the second projection and $\mathrm{pr}_1 : W \to A$ the first projection, so that

$$g \circ \pi = [p] \circ \mathrm{pr}_1 \,. \tag{4.5}$$

Since $[p]$ is finite étale of degree p^{2d} [Mum74, §6, Application 3 and §7, Theorem 4, Corollary 1], π is finite étale of degree p^{2d}. Thus W is projective. The group G acts on W by $h(b, z) := (b+h, z)$, since $[p](b + h) = [p](b)$. The G-action on W is free, since $b + h = b$ implies $h = 0$. Moreover, $\pi(hw) = \pi(w)$ and $\mathrm{pr}_1(hw) = \mathrm{pr}_1(w) + h$ for any $h \in G$ and $w \in W$, so pr_1 is G-equivariant.

Step 2. In this step we show that W is integral and that $\deg \mathrm{pr}_1 = r$.

Since π is étale and Z is integral, W is reduced [Gro67, Proposition 17.5.7]. Let W_1 be an irreducible component of W. The complement in W_1 of the other irreducible components of W is a non-empty open subset of W, and its image under π is open, since π is flat and of finite type [Har77, Chapter III, Exercise 9.1]. Since π is finite, $\pi(W_1)$ is closed. Since Z is irreducible, $\pi(W_1) = Z$. Thus $\pi|_{W_1} : W_1 \to Z$ is dominant generically finite. Since W is reduced and π is finite flat of degree p^{2d}, the sum of $\deg(\pi|_{W_1})$ over all irreducible components W_1 of W is p^{2d}. In particular, $\deg(\pi|_{W_1}) \le p^{2d}$. By (4.5), $[p](\mathrm{pr}_1(W_1)) = g(Z)$ is dense in A, and $\mathrm{pr}_1(W_1)$ is closed, so $\mathrm{pr}_1(W_1) = A$. Computing the degree of $g \circ \pi|_{W_1} = [p] \circ \mathrm{pr}_1 |_{W_1}$ in two ways, we obtain

$$r \cdot \deg(\pi|_{W_1}) = p^{2d} \cdot \deg(\mathrm{pr}_1 |_{W_1}).$$

Since $p \nmid r$, p^{2d} divides $\deg(\pi|_{W_1}) \le p^{2d}$. Hence $\deg(\pi|_{W_1}) = p^{2d}$, and so $W_1 = W$. Thus W is integral, and $\deg \mathrm{pr}_1 = r \cdot p^{2d}/p^{2d} = r$.

Step 3. In this step we send a point w of W over a general point of Z to the tuple of the images of hw, $h \in G$, in W', and take the closure T_1 of the image. We show that T_1 is an irreducible G-stable variety with $\dim T_1 < d$ and with a faithful G-action, and we obtain a G-equivariant dominant rational map $W \dashrightarrow T_1$.

Since $W' \to Z'$ is étale over a dense open subset of Z' (Definition 2.1), any irreducible component of W' which dominates Z', with the reduced structure, is finite over a dense open subset of Z' [Har77, Chapter II, Exercise 3.7]. Removing from Z' the closure of the image of the irreducible components of W' which do not dominate Z', we may assume, after shrinking Z' to a dense affine open subset, that $W' \to Z'$ is finite étale. Then W' is affine. There exists a dense open subset $Z^\circ \subset Z$ such that the rational map $Z \dashrightarrow Z'$ is a morphism $\psi : Z^\circ \to Z'$ on Z°, and the isomorphism fixed in **Step 1** extends to an isomorphism

$$\pi^{-1}(Z^\circ) \cong W' \times_{Z'} Z^\circ$$

over Z° [Gro66, Corollary 8.8.2.5]. Let $\lambda : \pi^{-1}(Z^\circ) \to W'$ be the composition of the isomorphism $\pi^{-1}(Z^\circ) \cong W' \times_{Z'} Z^\circ$ with the first projection. Then the image of $\lambda(w)$ in Z' is $\psi(\pi(w))$ for any $w \in \pi^{-1}(Z^\circ)$, and the morphism $w \to (\lambda(w), \pi(w))$ is injective on $\pi^{-1}(Z^\circ)$.

Since $\pi(hw) = \pi(w)$, the open subset $\pi^{-1}(Z^\circ)$ is G-stable. Let

$$\lambda_G : \pi^{-1}(Z^\circ) \to \prod_{h \in G} W', \quad w \to (\lambda(hw))_{h \in G},$$

and let G act on $\prod_{h \in G} W'$ by $h' \cdot (w_h)_{h \in G} := (w_{h+h'})_{h \in G}$. Then

$$\lambda_G(h'w) = (\lambda((h + h')w))_{h \in G} = h' \cdot \lambda_G(w),$$

so λ_G is G-equivariant. Let T_1 be the closure of the image of λ_G, with the reduced structure. Since W is integral, T_1 is irreducible, and T_1 is G-stable since λ_G is G-equivariant.

All coordinates of $\lambda_G(w)$ map to the point $\psi(\pi(w))$ of Z'. Thus T_1 is contained in the $|G|$-fold fiber product of W' over Z', which is closed in $\prod_{h\in G} W'$ and finite over Z'. Hence $\dim T_1 \le \dim Z' < d$.

We show that the G-action on T_1 is faithful. Assume that $h' \in G$ acts trivially on T_1. Then $\lambda_G(h'w) = \lambda_G(w)$ for any $w \in \pi^{-1}(Z^\circ)$, and the coordinate indexed by $0 \in G$ implies $\lambda(h'w) = \lambda(w)$. Since $\pi(h'w) = \pi(w)$ and $w \to (\lambda(w), \pi(w))$ is injective, $h'w = w$. Since the G-action on W is free, $h' = 0$.

Step 4. In this step we construct X, T, q, and f.

The scheme W' is a closed subscheme of an affine space $\mathbb{A}^{N'}$. Thus $\prod_{h\in G} W'$ is a closed subscheme of $(\mathbb{A}^{N'})^G \subset (\mathbb{P}^{N'})^G$, where G acts on $(\mathbb{P}^{N'})^G$ by $h' \cdot (t_h)_{h\in G} := (t_{h+h'})_{h\in G}$. Let $\bar{T}_1$ be the closure of T_1 in $(\mathbb{P}^{N'})^G$. Then $\bar{T}_1$ is a G-stable projective variety, T_1 is a dense open subset of $\bar{T}_1$, and the G-action on $\bar{T}_1$ is faithful. The hypotheses of [AW97, Theorem 0.1] hold for $\bar{T}_1$, the group G, and the closed subset $\mathrm{Sing}(\bar{T}_1)$: $\bar{T}_1$ is a projective variety, G is a finite group acting faithfully on $\bar{T}_1$, and $\mathrm{Sing}(\bar{T}_1)$ is a G-stable proper closed subset of $\bar{T}_1$, since $\bar{T}_1$ is a variety [Har77, Chapter II, Corollary 8.16]. Thus by [AW97, Theorem 0.1], there exists a G-equivariant modification $T \to \bar{T}_1$, that is, a G-equivariant proper birational morphism (see [AW97, §1]), such that T is a smooth projective variety. Since $T \to \bar{T}_1$ is surjective, the G-action on T is faithful, and $\dim T = \dim T_1 < d$.

The composition of λ_G with the inverse of the birational morphism $T \to \bar{T}_1$ is a G-equivariant dominant rational map $W \dashrightarrow T$. Let $\bar{W} \subset W \times T$ be the closure of the graph of $W \dashrightarrow T$. Then $\bar{W}$ is a projective variety which is stable under the diagonal G-action, and the first projection $\bar{W} \to W$ is a G-equivariant birational morphism. Since the G-action on W is free, the G-action on $\bar{W}$ is faithful. By [AW97, Theorem 0.1] for $\bar{W}$, the group G, and the G-stable closed subset $\mathrm{Sing}(\bar{W})$, which is a proper closed subset since $\bar{W}$ is a variety [Har77, Chapter II, Corollary 8.16], there exists a G-equivariant modification $X \to \bar{W}$ such that X is a smooth projective variety. Then $\dim X = \dim W = d$.

Let q be the composition

$$X \to \bar{W} \to W \xrightarrow{\mathrm{pr}_1} A,$$

and let f be the composition $X \to \bar{W} \to T$. Both are G-equivariant. Since $X \to \bar{W} \to W$ is birational and $\deg \mathrm{pr}_1 = r$ (**Step 2**), q is dominant generically finite of degree r, and $p \nmid r$. Since $W \dashrightarrow T$ is dominant and $\bar{W}$ is projective, $\bar{W} \to T$ is surjective. Thus f is surjective. □

Proof of Theorem 1.2. Let k be a field of characteristic zero, A an abelian variety of dimension d over k, and p a prime. The case $d = 0$ is trivial. Assume that $d \ge 1$, and suppose that $[p] : A \to A$ is not p-incompressible. By Lemma 4.1, there exists an abelian variety A' of dimension d over $\mathbb{C}$ such that $[p] : A' \to A'$ is not p-incompressible. By Lemma 4.2 applied to A' and $G := A'[p]$, there exist smooth projective varieties X and T with regular G-actions, a G-equivariant dominant generically finite morphism $q : X \to A'$ with $p \nmid \deg q$, and a G-equivariant surjective morphism $f : X \to T$, such that $\dim X = d$, $\dim T < d$, and the G-action on T is faithful. By Theorem 1.3 with $n := \dim T$, $p^{d-\dim T}$ divides $\deg q$. Since $d - \dim T \ge 1$, p divides $\deg q$, a contradiction. □

References

[AW97] D. Abramovich and J. Wang, *Equivariant resolution of singularities in characteristic 0*, Math. Res. Lett. **4** (1997), no. 3, 427–433.

[AB67] M. F. Atiyah and R. Bott, *A Lefschetz fixed point formula for elliptic complexes. I*, Ann. of Math. (2) **86** (1967), 374–407.

[AB68] M. F. Atiyah and R. Bott, *A Lefschetz fixed point formula for elliptic complexes. II. Applications*, Ann. of Math. (2) **88** (1968), 451–491.

[AM69] M. F. Atiyah and I. G. Macdonald, *Introduction to commutative algebra*, Addison-Wesley Publishing Co., Reading, Mass.–London–Don Mills, Ont. (1969).

[BFN10] D. Ben-Zvi, J. Francis, and D. Nadler, *Integral transforms and Drinfeld centers in derived algebraic geometry*, J. Amer. Math. Soc. **23** (2010), no. 4, 909–966.

[BF24] P. Brosnan and N. Fakhruddin, *Fixed points, local monodromy, and incompressibility of congruence covers*, J. Algebraic Geom. **33** (2024), no. 2, 295–346.
[BR97] J. Buhler and Z. Reichstein, *On the essential dimension of a finite group*, Compositio Math. **106** (1997), no. 2, 159–179.
[Che26] Q. Chen, *Lower bounds on essential dimension for congruence covers of mixed Shimura varieties*, arXiv:2605.25628 (2026).
[CT02] J.-L. Colliot-Thélène, *Exposant et indice d'algèbres simples centrales non ramifiées*, with an appendix by O. Gabber, Enseign. Math. (2) **48** (2002), 127–146.
[FS22] N. Fakhruddin and R. Saini, *Finite groups scheme actions and incompressibility of Galois covers: beyond the ordinary case*, Doc. Math. **27** (2022), 151–182.
[FKW21] B. Farb, M. Kisin, and J. Wolfson, *The essential dimension of congruence covers*, Compos. Math. **157** (2021), no. 11, 2407–2432.
[FKW24] B. Farb, M. Kisin, and J. Wolfson, *Essential dimension via prismatic cohomology*, Duke Math. J. **173** (2024), no. 15, 3059–3106.
[Gro65] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas. II*, Inst. Hautes Études Sci. Publ. Math. **24** (1965), 231 pp.
[Gro66] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas. III*, Inst. Hautes Études Sci. Publ. Math. **28** (1966), 255 pp.
[Gro67] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas. IV*, Inst. Hautes Études Sci. Publ. Math. **32** (1967), 361 pp.
[Har77] R. Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics, No. 52, Springer-Verlag, New York–Heidelberg (1977).
[KM08] N. A. Karpenko and A. S. Merkurjev, *Essential dimension of finite p-groups*, Invent. Math. **172** (2008), no. 3, 491–508.
[KZ26] J. Kollár and Z. Zhuang, *Essential dimension of isogenies*, Pure Appl. Math. Q. **22** (2026), no. 2, 675–690; arXiv:2402.15362.
[KP20] G. Kondyrev and A. Prikhodko, *Categorical proof of holomorphic Atiyah–Bott formula*, J. Inst. Math. Jussieu **19** (2020), no. 5, 1739–1763.
[LOZ25] Y. Luo, K. Oguiso, and D.-Q. Zhang, *Essential dimensions of polarized endomorphisms of abelian varieties*, arXiv:2512.04942 (2025).
[LZ26] Y. Luo and D.-Q. Zhang, *Galois self-covers of projective spaces and essential dimensions*, arXiv:2606.10207 (2026).
[Mat86] H. Matsumura, *Commutative ring theory*, Cambridge Studies in Advanced Mathematics **8**, Cambridge University Press, Cambridge (1986).
[Mum74] D. Mumford, *Abelian varieties*, Tata Institute of Fundamental Research Studies in Mathematics, No. 5, second edition, Oxford University Press, London (1974).
[RY00] Z. Reichstein and B. Youssin, *Essential dimensions of algebraic groups and a resolution theorem for G-varieties*, with an appendix by J. Kollár and E. Szabó, Canad. J. Math. **52** (2000), no. 5, 1018–1056.
[Ser77] J.-P. Serre, *Linear representations of finite groups*, Graduate Texts in Mathematics **42**, Springer-Verlag, New York–Heidelberg (1977).
[Sta] The Stacks Project Authors, *The Stacks project*, https://stacks.math.columbia.edu, 2026.

School of Mathematical Sciences, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Beijing International Center for Mathematical Research, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China
Email address: liujihao@math.pku.edu.cn